# Day 5 afternoon — Fit, predict, check

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Monday 16 November 2026 · Pontoppidanstræde 101, room 1.001

*Interpreting the fit · the mean versus one new observation · posterior predictive checks · revising the model · robust regression*

# Today

| | | |
|---|---|---|
| **08:45** | **Lecture A** | Regression as a generative model · scales · priors · prior predictive |
| **09:45** | **Exercise A** | Your own timing data: scales, priors, prior predictive · question set |
| **11:10** | **Peer review A** | Partner interviews · plenary · everyone back at 11:10 |
| **11:55** | **Lunch** | |
| **12:30** | **Lecture B** | Fit · mean vs new attempt · posterior predictive checks · robust |
| **13:30** | **Exercise B** | Fit, predict, check, revise the model · question set |
| **15:00** | **Peer review B** | Partner interviews · plenary · everyone back at 15:00 |
| **15:45** | **Daily diary** | Mentimeter: 3 things learned, 3 not yet familiar |

# Follow-up from the morning plenary

**Open points from Peer review A**

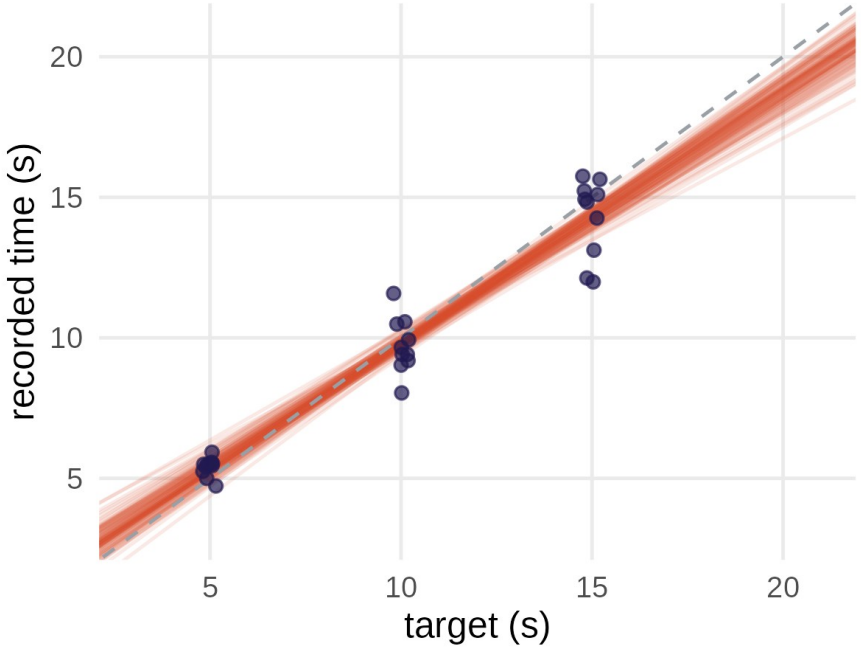
Fill in during lunch: the one or two questions the plenary did not settle.

1. …
2. …

# By the end of this lecture you can

1. **Interpret** the fitted regression with probabilities about β
2. **Separate** the mean at a new x from one new observation there
3. **Check** the model with posterior predictive checks
4. **Revise** the model when a check fails, and check again
5. **Protect** the fit against an unusual observation (Student-t errors)

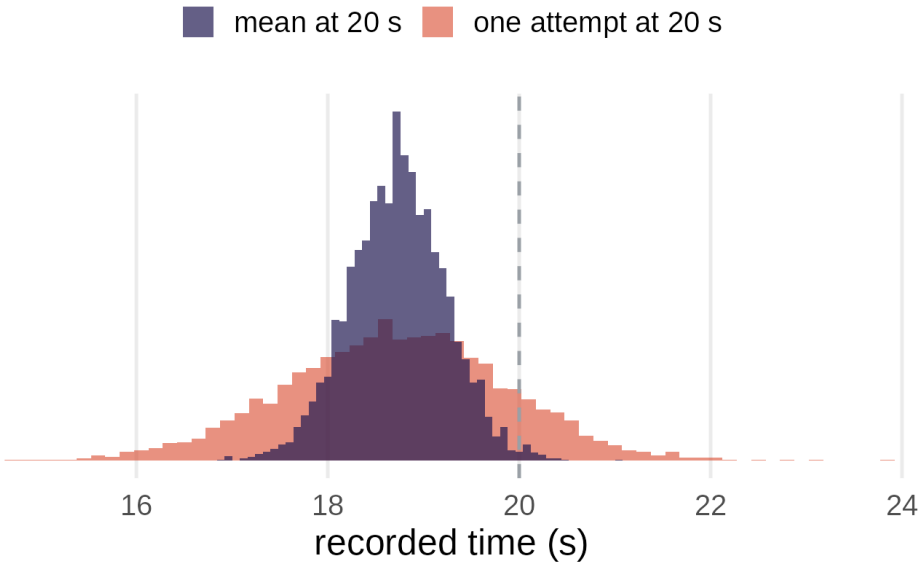
# The fit




| | mean | 90% interval |
|---|---|---|
| α (s) | 9.81 | 9.51 – 10.10 |
| β | 0.89 | 0.82 – 0.97 |
| σ (s) | 1.03 | 0.82 – 1.29 |

- No divergences, R-hat 1.00, ESS > 3,000.
- Each red line: $\alpha^{(s)} + \beta^{(s)} \cdot x$ for one draw.
- $P(\beta < 1 \mid y) = 0.99$: her clock very probably compresses.

# At 20 s: the mean versus one attempt




**Mean at 20 s:**

$\mu_{20}^{(s)} = \alpha^{(s)} + \beta^{(s)} \cdot 10$

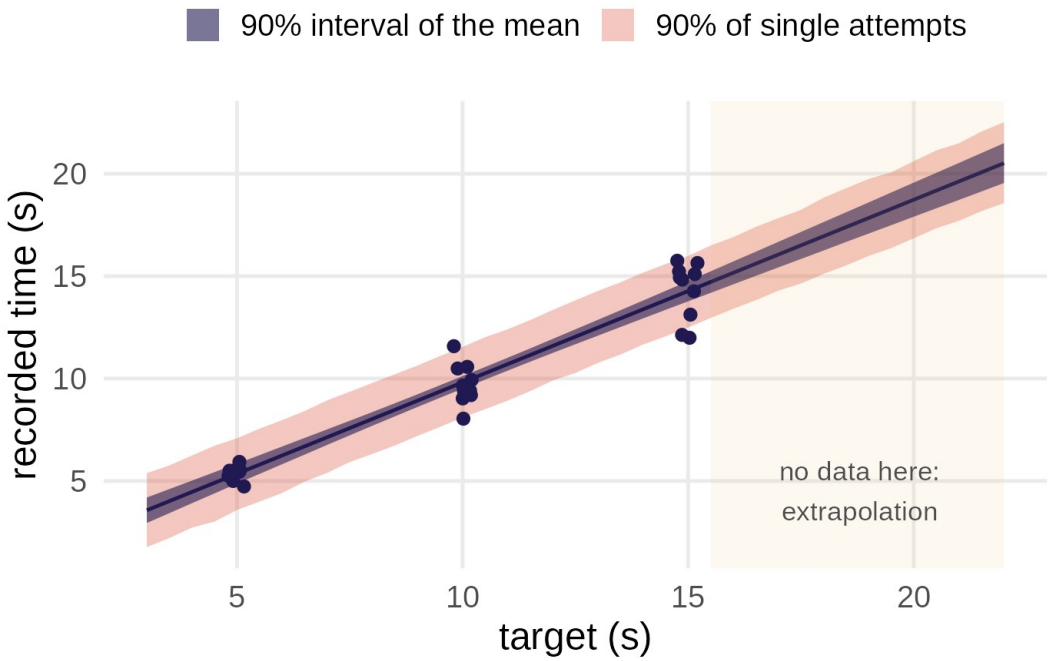
**One attempt at 20 s:**

$\tilde{y}_{20}^{(s)} \sim \text{Normal}(\mu_{20}^{(s)}, \sigma^{(s)})$

| | 90% interval | P(< 20 s) |
|---|---|---|
| mean | 17.9 – 19.6 | 0.99 |
| one attempt | 16.8 – 20.5 | 0.87 |

*Same idea as Day 4: an average is not one more attempt. And 20 s was never tried.*

# Prediction across all targets




- Dark band: where the mean line is (uncertainty about α and β).
- Light band: where 90% of single attempts fall (adds σ).
- Beyond 15 s the bands rest on the assumption that the line stays straight: extrapolation needs a reason, not just a model.

# Posterior predictive checks

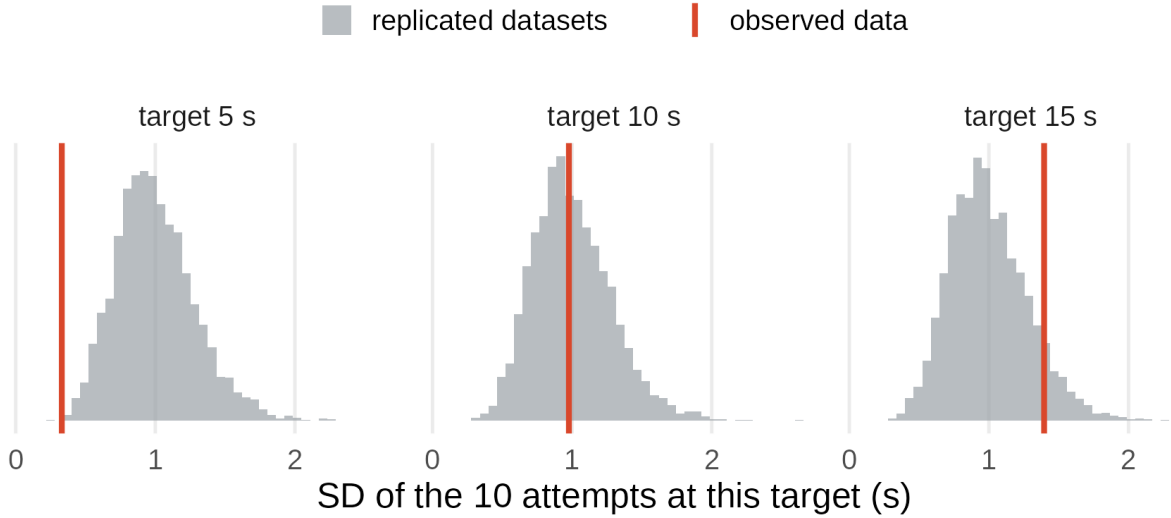
**The question**

If the model were right, would data like ours be typical?

1. For each draw, simulate a new dataset y_rep at the same 30 targets (generated quantities).
2. Choose a statistic T that matters for the question: here, the SD of the attempts at each target.
3. Compare T(observed data) with the distribution of T(y_rep).

```
array[N] real y_rep = normal_rng(alpha + beta * x, sigma);
```

# A check: the spread at each target

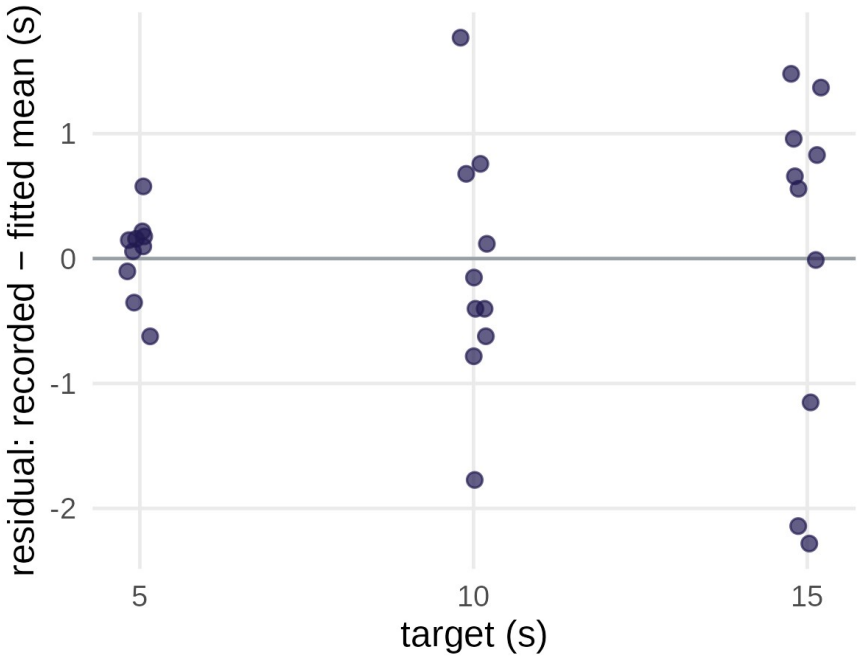



| target | observed SD | replicated 90% |
|---|---|---|
| 5 s | 0.33 | 0.58 – 1.50 |
| 10 s | 0.98 | 0.59 – 1.49 |
| 15 s | 1.40 | 0.59 – 1.50 |

At 5 s her attempts are far less spread than any replicated dataset; at 15 s more spread than most. The model's single σ misses that the spread grows with the target.

*The check found what a summary of α, β and σ cannot show.*

# The residuals show it too



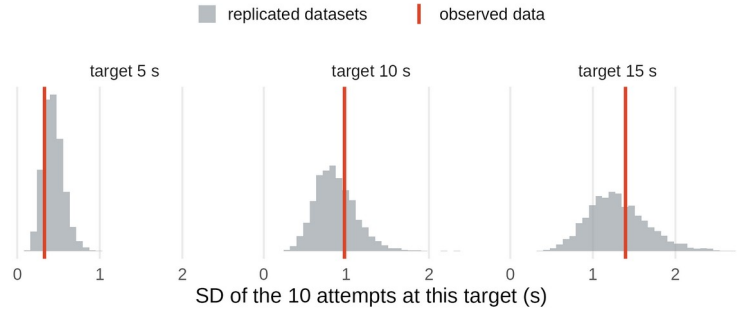

- Residual: recorded time − fitted mean.
- At 5 s within ± 0.6 s; at 15 s up to ± 2.3 s: a funnel.
- Known from timing research: the error grows with the duration (scalar timing).
- A model check should look for exactly such patterns.

# Revise: the spread grows with the target

```
model {
  ...
  // spread at target t is sigma * t / 10
  y ~ normal(alpha + beta * x, sigma * (x + 10) / 10);
}
```

One changed line: σ is now the spread at a 10-s target; it halves at 5 s and doubles at 20 s.

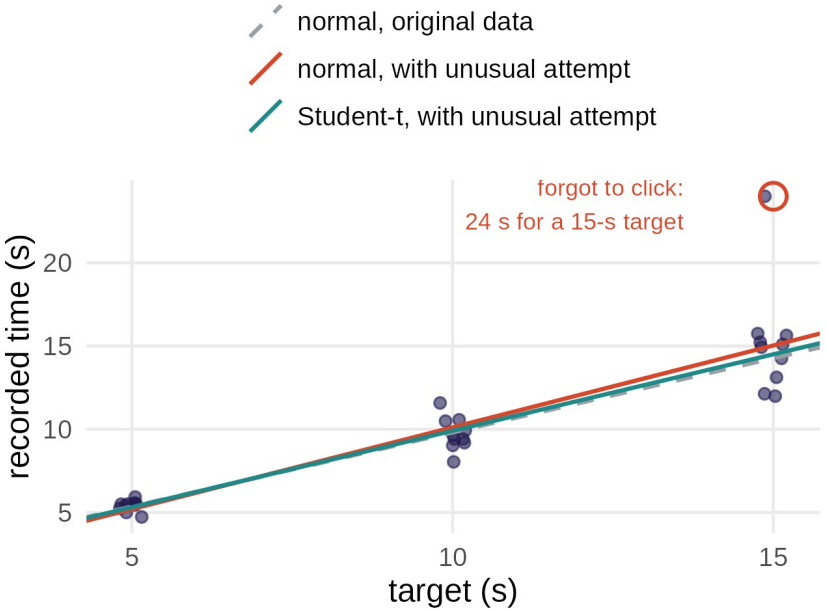
| | constant σ | growing σ |
|---|---|---|
| β | 0.89 | 0.88 |
| one attempt at 20 s | 16.8 – 20.5 | 15.5 – 21.8 |
| SD check at 5 s | fails | passes |




The same check for the revised model: the observed SDs (red) are now typical.

- The slope hardly changes.
- The prediction at 20 s becomes wider — and more honest.

# An unusual attempt




- Suppose one 15-s attempt was recorded as 24 s: she forgot to click.
- Normal model: β 0.89 → 0.98, σ 1.03 → 2.08. One attempt moves both.
- Is it an error (remove it, and say so) or a real, rare event (keep it, and use a model that expects such events)?

# Robust regression: Student-t errors

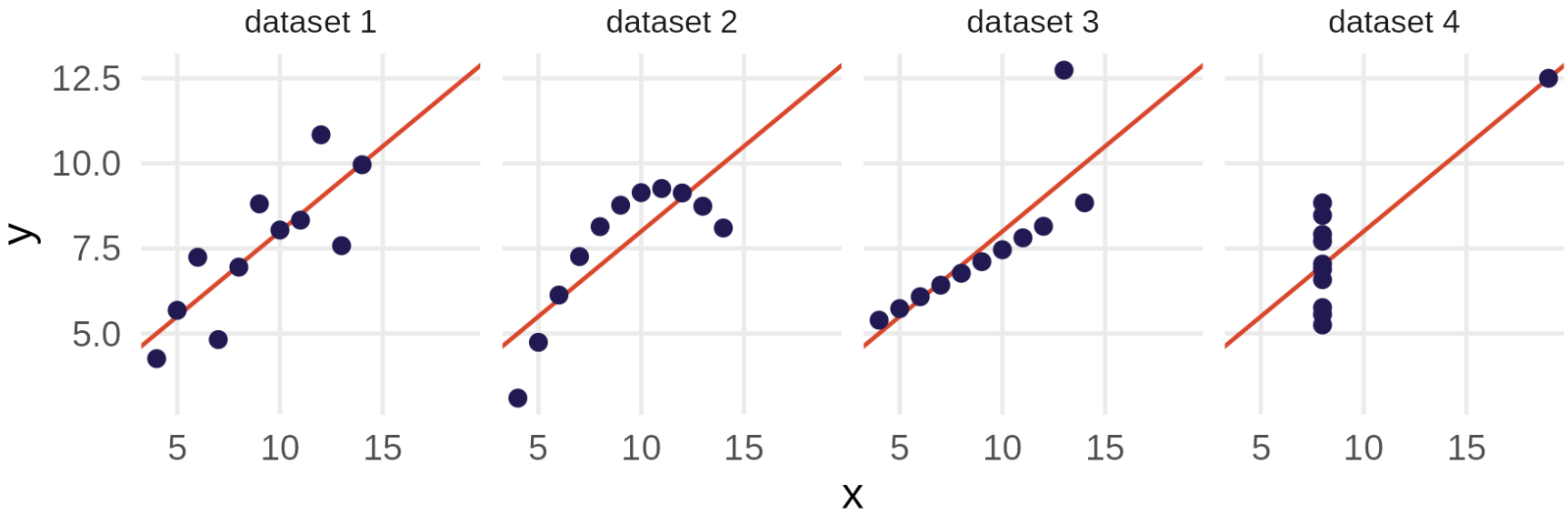
```
y ~ student_t(4, alpha + beta * x, sigma);
```

- The t distribution with 4 degrees of freedom has heavier tails: a large residual is less surprising, so one attempt pulls less.
- σ is now a scale, not the SD of the attempts.

| fit | β | 90% interval |
|---|---|---|
| normal, original data | 0.89 | 0.82 – 0.97 |
| normal, with 24-s attempt | 0.98 | 0.84 – 1.13 |
| Student-t, with 24-s attempt | 0.92 | 0.83 – 1.00 |

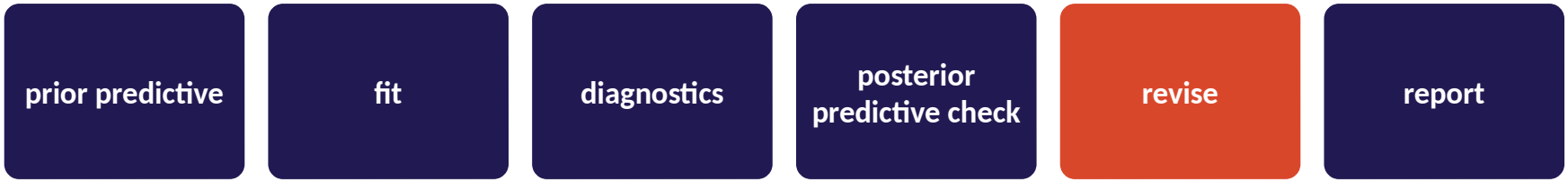
*A robust model is a modelling choice, not a way to make inconvenient data disappear.*

# Always plot: Anscombe's quartet




All four datasets: mean of y 7.50, slope 0.50, correlation 0.82. The same regression line; four different stories.

# The modelling workflow




- Prior predictive: does the model expect plausible data before seeing any? (morning)
- Diagnostics: did we compute the posterior correctly? (Day 3)
- Posterior predictive check: does the fitted model reproduce what matters in the data?
- Revise and check again; report the model you used and why.

# Worked example: checks and predictions

### Question 1

Why is the 90% interval for one attempt at 20 s wider under the growing-spread model (15.5–21.8) than under constant spread (16.8–20.5)?

### Question 2

At 10 s the observed SD (0.98) lies in the middle of the replicated values. Does that prove the constant-spread model is right?

### Question 3

Another person's data contain one 24-s attempt at the 15-s target. List two things you would do before choosing between removing it and a Student-t model.

# Worked example: answers

## Question 1

Under growing spread, σ at 20 s is twice σ at 10 s (about 1.8 s instead of 1.0 s). The data show that the spread grows, so the wider interval is the honest one; the constant model borrows the average spread from all targets.

## Question 2

No. A check can only show a misfit in the statistic it looks at. At 10 s the average spread happens to fit; at 5 and 15 s it does not. Choose statistics that matter for the question, and look at several.

## Question 3

Check the record (was the session interrupted? is it a typo?) and ask the person if possible; refit with and without it and report whether conclusions change. Never delete it silently.

# Exercise B · 13:30 – 15:00

- Fit the regression in Stan and check the diagnostics.
- P(β < 1); the mean versus one attempt at 20 s.
- Posterior predictive check: the spread at each target.
- Revise the model if the check fails, and check again.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

**15:00**

peer review B

If Stan does not compile on your laptop: continue on UCloud or a group member’s laptop. New roles this block.

# Peer review B · 15:00 – 15:45

## 20 min

### Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

## 23 min

### Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

## 2 min

### Wrap-up

Points that need longer go to tomorrow's follow-up.

**Partner pairs:** 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

# Daily diary · 15:45 – 16:00

**menti.com · code: ____ ____**

1. Three things I have learned today
2. Three things I am not yet very familiar with